

Smart Punch-In System

Chuka Orajuba

BEng Software & Electronic Engineering

PROJECT SUMMARY

This project presents a fully integrated, biometric attendance management system built on a Raspberry Pi 4. The system uses a camera module and the face_recognition library to identify employees in real time, automatically logging clock-in and clock-out events via REST API. All attendance records, employee profiles, shift rosters, and time entries are stored locally in a SQLite database running directly on the Pi, keeping the system self contained without requiring an external server. A React web dashboard provides employees and administrators with live attendance data, shift schedules, and attendance reports.

HARDWARE

- Raspberry Pi 4 Model B (4GB RAM)
- Raspberry Pi Camera Module v2
- Local SQLite database (on Raspberry Pi)
- Connected via home WiFi network
- Runs FastAPI backend and face recognition script simultaneously

SOFTWARE & LIBRARIES

BACKEND

- FastAPI (Python)
- SQLite Database
- Firebase Admin SDK
- Uvicorn ASGI Server
- face_recognition

FRONTEND

- React / Next.js
- Firebase Auth
- Server-Sent Events
- JavaScript

RECOGNITION

- OpenCV
- Picamera2
- dlib (liveness detection)
- face_recognition lib
- HOG face detection

SECURITY

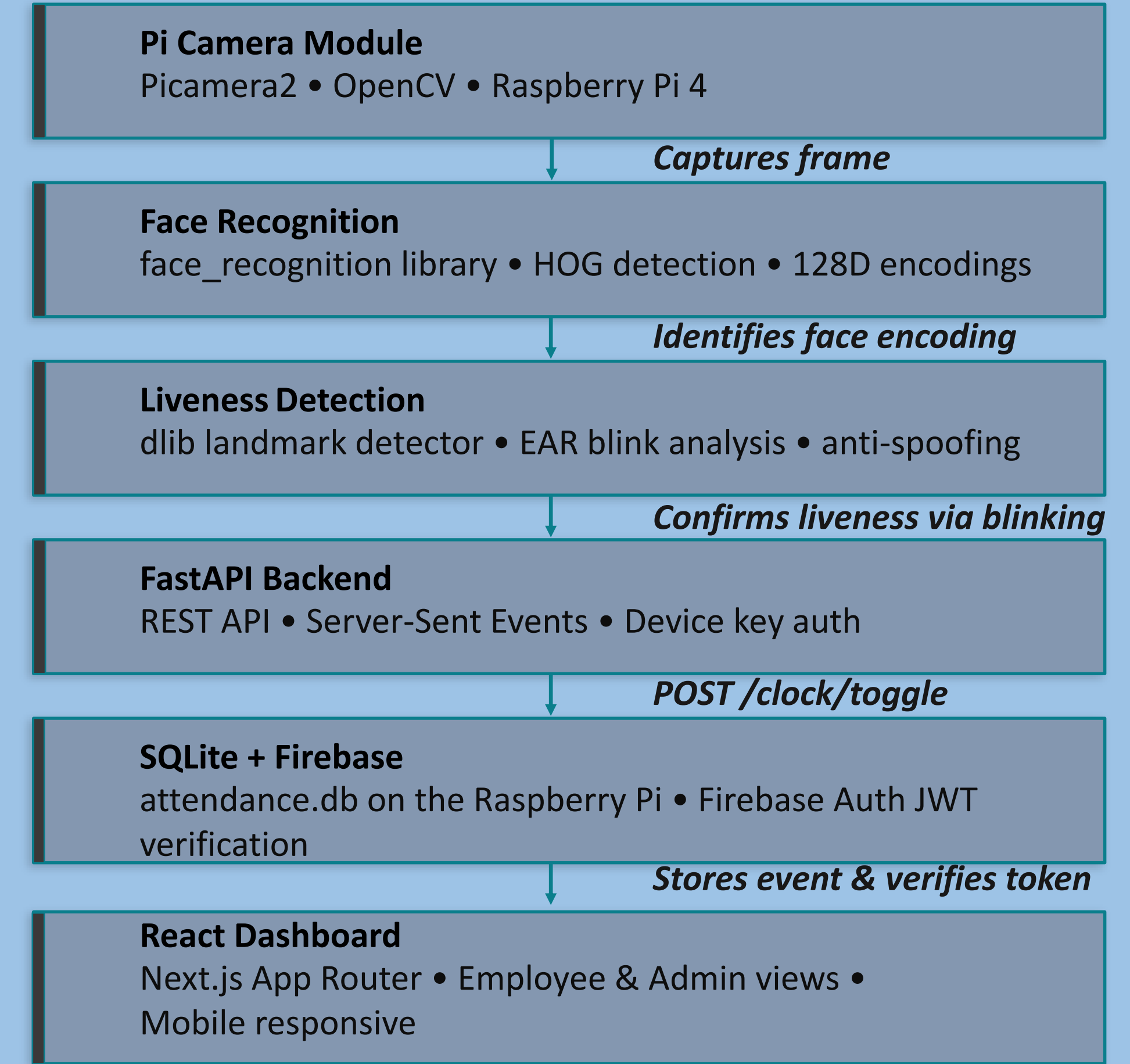
- Firebase JWT tokens
- Device API key
- CORS protection
- secrets.compare_digest

GITHUB – RASPBERRY PI

GITHUB – REACT WEBPAGE



BLOCK ARCHITECTURE



KEY FEATURES

Real-Time Face Recognition

Identifies employees via Pi Camera using HOG detection and 128D face encodings. Supports multiple faces simultaneously with a 10 second cooldown per person logic.

Liveness Detection

Prevents photo spoofing using Eye Aspect Ratio (EAR) blink detection via dlib. Clock events only fire after a confirmed natural blink is detected.

Live Dashboard Updates

Server-Sent Events push clock-in/out events to all connected browsers instantly

Admin Roster & Scheduling

Admin can build weekly shift rosters, assign shifts per employee per day, and view a full weekly calendar grid with colour coded shift types.

Attendance Report

Compares actual clock times against scheduled shifts. Automatically flags late arrivals, early departures, and no-shows with a configurable grace period.

Employee Registration

Admin selects a Pi-recognised employee from a dropdown, enters their email, and sends a Firebase password invite linking their face identity to their login account.

RESULTS

Recognition Performance:

100% recognition accuracy was achieved across all test attempts with no false positives recorded. Recognition was instant, with both speed and accuracy improving directly with the size of the training dataset.

Liveness Detection:

Blink based liveness detection reliably blocked all photo spoofing attempts during testing. Detection was quick, with response time also benefiting from a larger training dataset.

Confidence Scores:

Face distance scores were consistently below 0.65 across all successful recognitions, with scores decreasing as more training images were added, confirming that training data quantity is the primary driver of system performance.

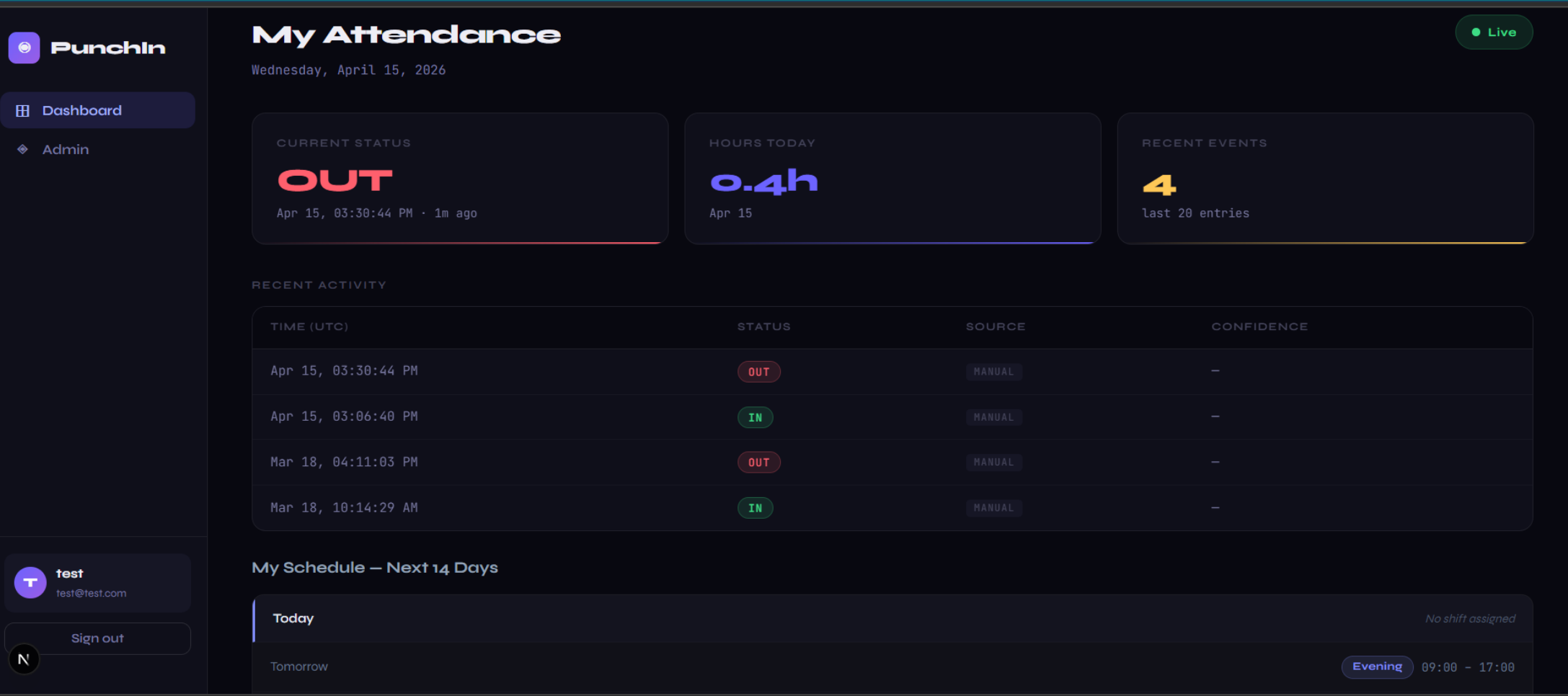
API Response Time:

The FastAPI backend responded to all clock events instantly on the local network, with no noticeable delay observed during testing.

APPLICATIONS

- Small to medium workplace attendance tracking
- Replacement for RFID and PIN systems
- Shift based environments such as retail, hospitality, healthcare
- Secure areas requiring liveness verified access logging

DASHBOARD SCREENSHOT



CONCLUSIONS

The project successfully demonstrates a production ready biometric attendance platform. Face recognition combined with liveness detection provides a secure and contactless clock in/out mechanism. The full-stack architecture from embedded Pi hardware through to a responsive webpage shows the integration of multiple disciplines including backend API design, database management, and frontend development.